

# 0005 What is Node.js?

Saturday, 24 June 2023

6:47 AM

- Node.js is an open source, cross-platform JavaScript Runtime Environment.
- OpenSource:- Source code is publicly available for sharing & modification.
- Cross-platforms:- available on macOS, Windows, Linux etc.
- JavaScript Runtime Environment:- Provides all the necessary components in order to use & run a JavaScript program Outside the browser.

↑  
node.js can run JS program outside browser.

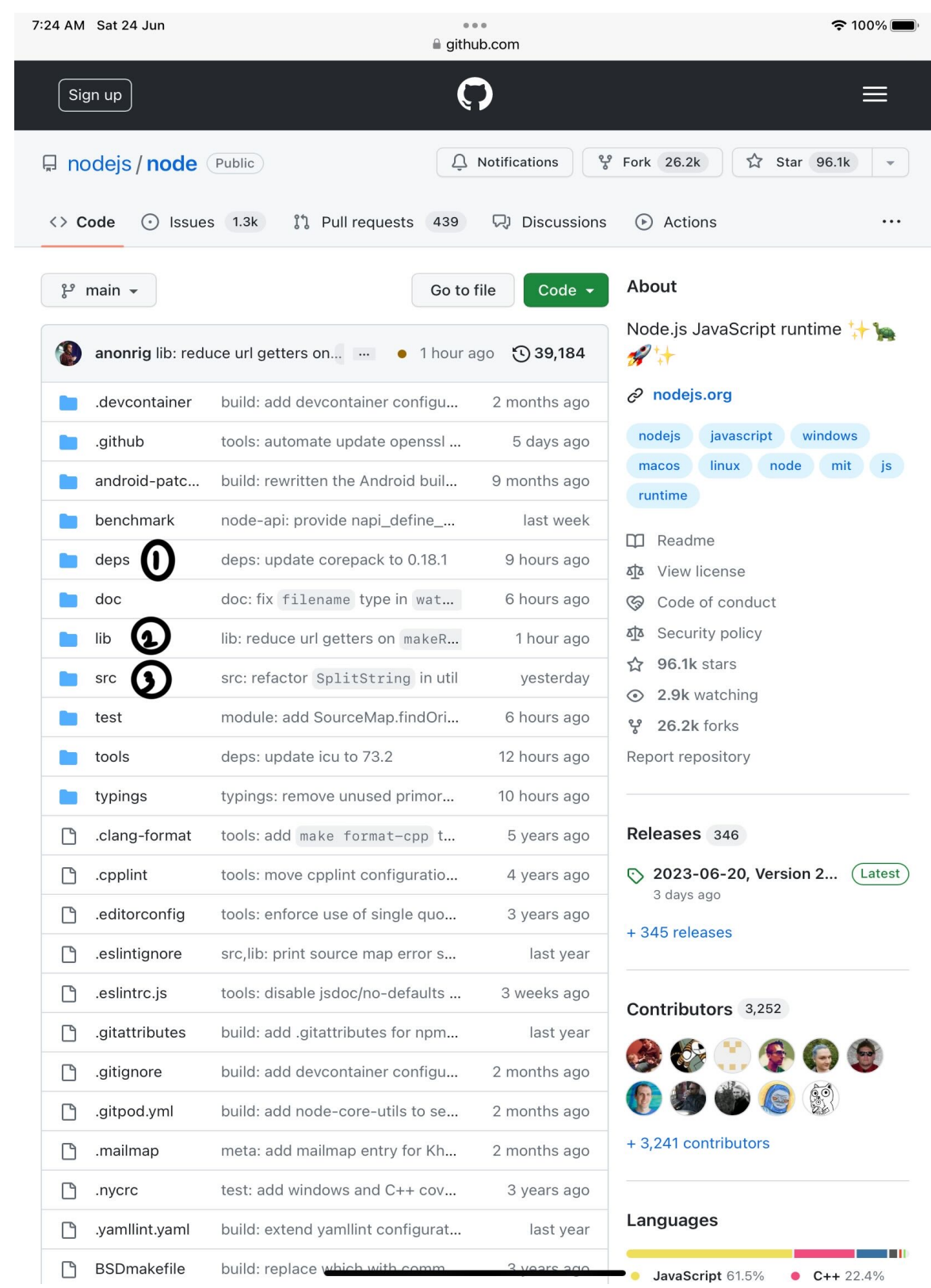
## Q. What can you build with Node.js?

- Before 2009 - where Node.js is introduced JS can only be run inside a browser.
- Using Node.js we can run JS outside browser.
  - We can build
    - Traditional Websites
    - Back-end services like APIs
    - Real time Applications.
    - Streaming services like Netflix..
    - CLI - Command line Interface tools
    - Multiplayer games etc...
- In short Node.js allows you to build complex & powerful Application.

→ [github.com/nodejs/node](https://github.com/nodejs/node)

:- Contains source code of Node.js.

— 3 main Component - About which you have to know something.



### ① Deps:- Dependencies

This folder contains external codes that node.js requires for its functioning.

— There are so many dependencies .. from which 2 are mainly important Once.

① V8:- Same JS engine present in chrome browser.

— Without V8 there is no way to Node.js to understand JS code.

② UV:- LibUV:- openSource library:-

providing node JS access to the underline OS features such as FS: file System & n/w: networking.

2<sup>nd</sup> folders:- Src :- Source folder :-

src:- Contains C++ source code of the node-JS runtime.

— Keep in mind:- JS as language was not design to deal with low level functionality like FS & n/w.

— C++ was designed to do that.

∴ Src folder is a set of C++ features that Node.js use for FS handling & n/w operations.

3<sup>rd</sup> Folder:- Lib:- Library folder

— We know that Node.js adds new features to JS through C++ code.

— Q. How does JS developer can access these new C++ features?

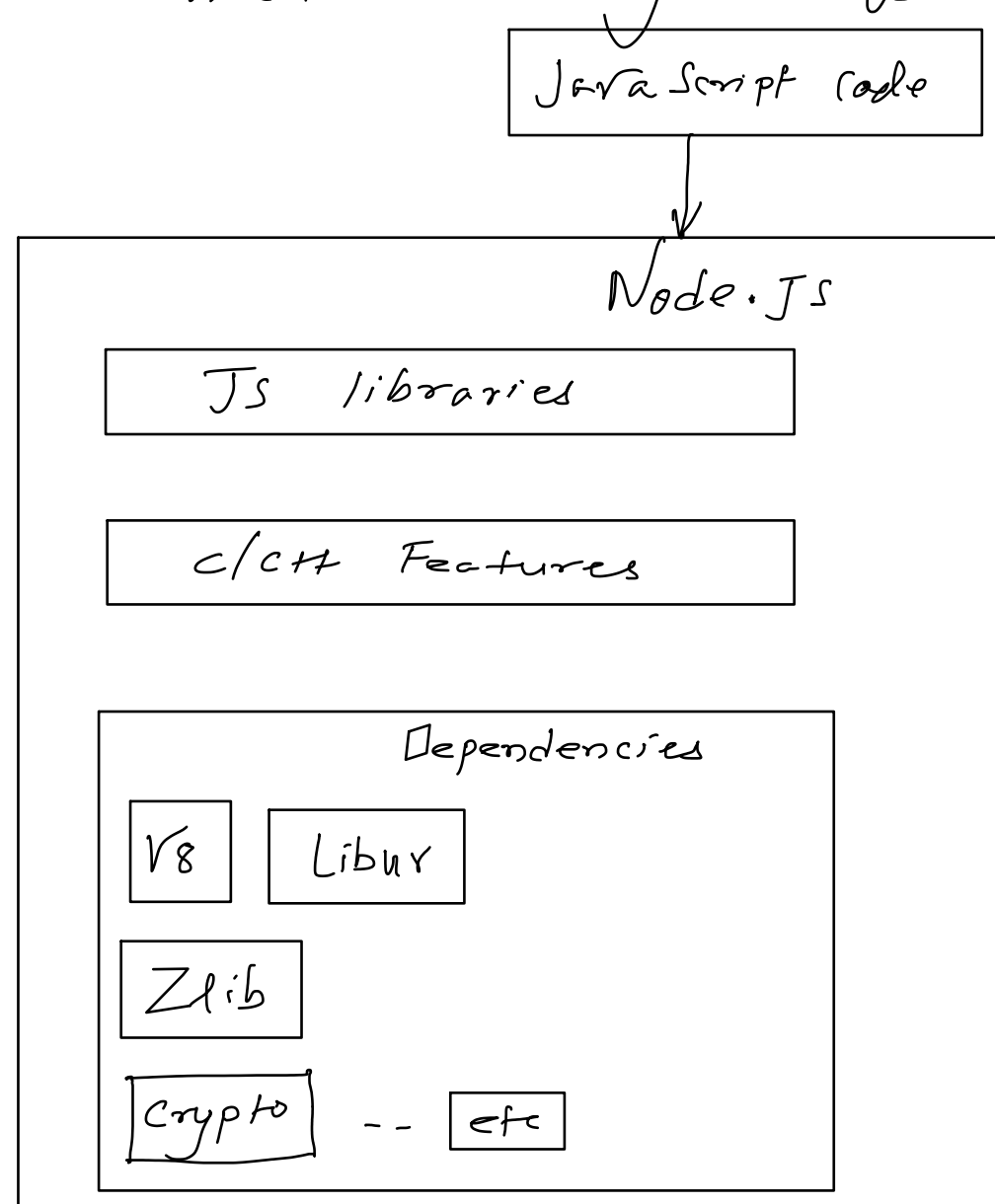
— Through Lib folder & its functionality.

— lib folder contains JS code that's been written for w that can easily access C++ features.

eg. fs.js - contains JS code we can write to access the file system.

— The code internally called the corresponding C++ features that relied on "libuv" to access the OS fns.

In fact lib folder also contains few utilities f that we need in writing Node.js Application.



Node.js Runtime

Node.js is code written in C++ & JS.

— Node.js Runtime Does not have access the Web API

— There is No Web Windows & No Documents in working with Node.js as like in web/browser based JS program.

→ Summary:-

- Node.js is open source, cross platform JS runtime Environment.
- It is Not language,
- It is Not Framework.
- It is Runtime &
- Capable of executing JS code outside browser
- It can executes Not only the standard ECMAScript language but also new features that are made available through C++ bindings using the V8 engine.
- It consists of C++ files which form the core features & JS files which expose common utilities & some of the C++ features for easier consumption.